2025秋高数上期中试题（回忆版）

1.单选

1. Equation $x^4 - x - 1 = 0$ has one root on the interval

(A) $\left(0, \frac{1}{2}\right)$

(B) $\left(\frac{1}{2}, 1\right)$

(C) $(1, 2)$

(D) $(2, 3)$

2. Let $f(0) = 0$. Which of the following conditions ensures that $f'(0)$ exists?

(A) $\lim\limits_{h \to 0} \dfrac{f\left(1 - \sqrt{1 - h^2}\right)}{h^2}$ exists.

(B) $\lim\limits_{h \to 0} \dfrac{f(2h) - f(h)}{h}$ exists.

(C) $\lim\limits_{h \to 0} \dfrac{f(h) - f(-h)}{2h}$ exists.

(D) $\lim\limits_{h \to 0} \dfrac{f(h^2 - h)}{h}$ exists.

3. Assume that $y = f(x)$ and $y = \sin x$ have the same tangent line at the origin. Then

$$\lim_{x \to +\infty} \left(2\mathrm{xf}\left(\frac{1}{3x} \right) \right) =$$

(A) $\frac{1}{36}$

(B) $\frac{4}{9}$

(C) $\frac{9}{4}$

(D) 36

4. Suppose $f(x) > 0$, $f'(x) < 0$, and $f''(x) > 0$ on $[a, b]$. Let

$$A_1 = \int_a^b f(x)\ \mathrm{dx}, A_2 = f(b)(b - a), A_3 = \frac{1}{2}[f(a) + f(b)](b - a)$$

Then
(A) $A_1 < A_2 < A_3$
(B) $A_2 < A_1 < A_3$
(C) $A_3 < A_1 < A_2$
(D) $A_2 < A_3 < A_1$

5.

Suppose $f(x)$ is differentiable at $x = a$, and $|f(x)|$ is not differentiable at $x = a$. Then

(A) $f(a) = 0,\ f'(a) = 0$
(B) $f(a) = 0,\ f'(a) \neq 0$
(C) $f(a) \neq 0,\ f'(a) = 0$
(D) $f(a) \neq 0,\ f'(a) \neq 0$

2.填空

(1) $f(x) = \int_1^x \left(2 - \frac{1}{t^{1/2}}\right)\mathrm{dt}\ (x > 0)$ is decreasing on the open interval ____.

(2) The linearization of $f(x) = (1-x)^{-5}(x+1)^{1/2}$ at $x = 0$ is $L(x) =$ _____.

(3) The asymptotes of the graph of the function $f(x) = \frac{x^2-1}{x} + \mathrm{xsin}\frac{1}{x}$ are _____.

(4) Given that $f(1) = 8$ and $f'(1) = 3$. If

$$h(x) = \sqrt[3]{\mathrm{xf}(x^2)},$$

then $h'(1) =$ _____.

(5) $f(x)$ is continuous on $(0, \infty)$, and

$$\int_0^{x^2(1+x)} f(t)\ \mathrm{dt} = x.$$

Then $f(2) =$ _____.

3.

Compute the first derivative of the following functions:

(1) $y = \sin\sqrt{1+\sqrt{x}}$

(2) $y = \frac{x^3+1}{1-x^2}$

4.

Find all non-differentiable points of the function

$$f(x) = x^2|\,x^3 - x|$$

on R.

5.

If $\sin x + \mathrm{xy} + y^2 = 1$, find the value of $\frac{d^2y}{\mathrm{dx}^2}$ at the point $(0, 1)$.

6.

Let $f(x) = \frac{x}{1+x^2}$.

(1) Identify where the local extrema of f occur. Find the function's local extreme values.
(2) Find the open intervals where the graph of f is concave up and where it is concave down.
(3) Sketch the graph.

7.

Find the constants a and b such that the function

$$f(x)=\begin{cases}\dfrac{\sin^2(3x)}{x^2+4x^3}, & x>0,\\ a, & x=0,\\ \dfrac{\sqrt{|1+2x|}-1-x}{\mathrm{bx}^2}, & x<0\end{cases}$$

is continuous at $x=0$.

8.

The equation

$$x^5-5x+k=0$$

has three distinct real roots. Find the range of k.